

Analysis

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Figure 1: Math tech, autumn tea

0 Real Number

§1. Rational Number Field ($\mathbb{Q}$)

Definition 0.1. A number r is called a **rational number**, if $r = \frac{p}{q}$, where $p, q \in \mathbb{N}$ and $q \neq 0$.

The set of all rational numbers is denoted by $\mathbb{Q}$, and it is called the **rational number field**.

Note. 1°. Proof by contradiction. – To verify if a number is irrational.

Properties. Basic properties of rational numbers: